Geometrical - Calculator

Project Proposal :

Geometric Calculator

Introduction:

This project aims to develop a user-friendly geometric calculator that allows users to calculate properties of 2D and 3D shapes. Users will be prompted to choose between 2D and 3D shapes.

Objective

Menu System
Develop a user-friendly menu system for users to choose between 2D and 3D shapes.

2D Shapes
Implement functions for calculating the area and circumference of 2D shapes (e.g., Circle, Rectangle).

3D Shapes
Implement functions for calculating the surface area and volume of 3D shapes (e.g.Sphere).

User Experience
Provide a seamless user experience with clear prompts and options.

Presentator : Umair

Working

1. Selection Users choose between 2D and 3D shapes in the main menu.	2. 2D Shapes 2 Sub-menu options for calculations (e.g., area, circumference) based on shape selection.	3. Input Users input the required parameters for the chosen calculation.
4. Calculation The program performs the calculation using the appropriate function.	5. Result The calculated result is displayed to the user.	6. Continuation 6 Users are prompted if they want to continue or exit the program

Analysis

1. User-Friendly Interface

The menu system ensures ease of use, guiding users through the shape and calculation selection process.

2. Modular Function Design

Functions are modular and focused on specific calculations for maintainability and extensibility.

Functions to Implement

• .calculateCircleArea(radius	• calculateCircleCircumference(radius)
• calculateRectangleArea(length,width)	• calculateRectanglePerimeter(length, width)
• calculateTriangleArea(base,height)	• calculateTrianglePerimeter(side1,side2, side3)

Presentator : umair

Functionality and Implementation

Structure and code the program to provide accurate calculations for each shape. Utilize both front-end and back-end technologies to create a seamless experience.

Conclusion

1. ***Enhanced Problem Solving***
   The geometric calculator promotes critical thinking and mathematical skills through practical application

2. ***Improved Efficiency***
   Streamline the calculation process, enabling users to quickly obtain accurate results..

Inspiration from Geometry

Geometry is not just about calculations, but also an art form. Explore the beauty and creativity of geometric shapes and structures.

Thanks . .